

GTIIT Spring 2025 Calculus 1M

Workshop 4 - Tuesday

March 24, 2025

Try to solve the following exercises. Feel free to chat about them with your classmates, but make sure to write your own solutions. If you get stuck, just move on to the next one and come back later.

If you need any help at all —whether it's a tiny hint or a big clue— don't hesitate to ask me!

Important note: If you find any errors or typos, please let me know so I can correct them.

1. Prove the statement using ε and δ definition of a limit and illustrate it with a graph:

(a) $\lim_{x \rightarrow 2} 2x - 2 = 2$

(b) $\lim_{x \rightarrow 4} -7 + \frac{1}{2}x = -5$

2. Prove the statement using ε and δ definition:

(a) $\lim_{x \rightarrow 3} x^2 = 9$

(b) $\lim_{x \rightarrow 789} c = c$, where c is some constant.

3. Suppose that $\lim_{x \rightarrow a} f(x) = +\infty$ and $\lim_{x \rightarrow a} g(x) = c$, where c is a real number. Prove each statement.

(a) $\lim_{x \rightarrow a} (f(x) + g(x)) = +\infty$,

(b) $\lim_{x \rightarrow a} (f(x)g(x)) = +\infty$ if $c > 0$,

(c) $\lim_{x \rightarrow a} (f(x)g(x)) = -\infty$ if $c < 0$.

4. Let $f(x) = 1/x$ and $g(x) = 1/x^2$.

(a) Compute $(f \circ g)(x)$

(b) is $f \circ g$ continuous everywhere?

5. Give an example of a function $f : \mathbb{R} \rightarrow \mathbb{R}$ such that f is continuous everywhere but in 0 and $|f|$ is continuous everywhere.

6. Use the definition of continuity and the properties of limits to show that the function

$$f(x) = \frac{x^2 + 5x}{2x + 1}$$

is continuous at 2.

7. Find the values of c that make the function f continuous everywhere.

$$f(x) = \begin{cases} cx^2 + 2x & \text{if } x < 2 \\ x^3 - cx & \text{if } x \geq 2. \end{cases}$$

8. Compute the limits:

(a) $\lim_{x \rightarrow \infty} \frac{x^2}{\sqrt{x^4+1}}$

(b) $\lim_{x \rightarrow \infty} \frac{x+3x^2}{4x-1}$

9. Use the squeeze Theorem to evaluate the limit $\lim_{x \rightarrow \infty} \frac{\sin x}{x}$.